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MATHEMATICS [CLASS 8th]

CHAPTER - 2

LINEAR EQUATION IN ONE VARIABLE

(highest power in linear equation is 1)

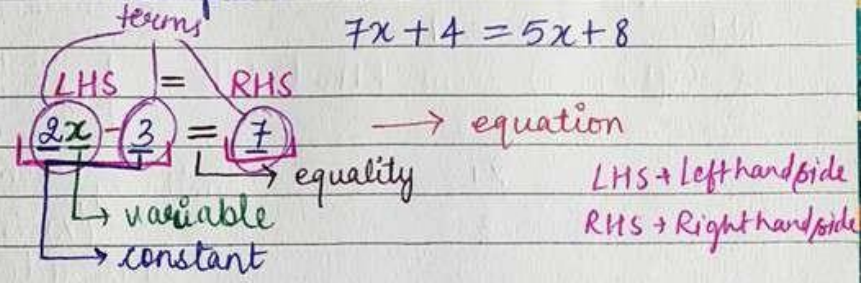
Notes :

Equation is a statement that supports the equality of two expressions, which are connected by equal sign (=)

constant - a value or number that never changes.
variable - an unknown value represented as a letter

linear equation in one variable has only a single variable

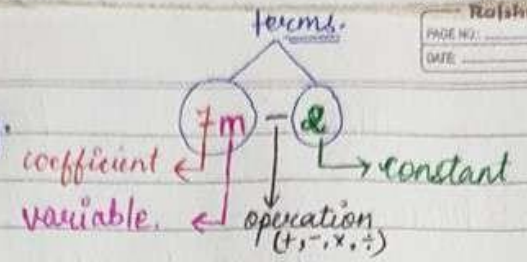
example : $7x+8=29$
$7x+4=5x+8$



SOLVING EQUATIONS HAVING THE VARIABLE ON BOTH SIDES

Equations can have variables on both sides.
In this type of problem, we have to bring all the consonents on one side and all the terms having variable on the other side. This way it can be

solved easily.



example :- Let's solve two questions to see how we can solve variables on both sides.

(a) $2x - 3 = 6 - x$ [we have variables on both sides [LHS & RHS]]

on shifting (-3) from LHS to RHS

$2x = 6 - x + 3$ [when changing position of any term the signs will change too]

on shifting variable $(-x)$ from RHS to LHS.

$2 \times x$

$2x + x = 6 + 3$ [Now we have variables on one side and constant on another side]

Add :- $3 \times x$

$3x = 9$

on shifting (3) from RHS to LHS

so $x = \frac{9}{3}$ 3 1

$x = 3$

[when, (+) is shifted to other side it becomes (−) similarly (×) is shifted to other side it becomes (÷) on vice versa in all cases]

(b) $9(t - 5) = 4t + 5$ [multiply 9 with both t and −5 to solve brackets]

$9t - 45 = 4t + 5$

$9t - 4t = 5 + 45$ [shifted all constant one side and variable on the other side]

$$5t = 50$$

$$t = \frac{\cancel{50}^{10}}{\cancel{5}}$$

[5 was in multiply with (t) on shifting to RHS from LHS it will be in divide]

$$t = 10$$

REDUSING EQUATIONS TO SIMPLER FORM.

We will take LCM and will solve the remaining equation from above method. Let's see some examples to solve these complex equation.

Ex - (a) $\frac{x}{2} - \frac{1}{5} = \frac{x}{3} + \frac{1}{4} + 1$

First shift all variable on one side and constant on the other side then take LCM.

$$\frac{x^{\times 3}}{2^{\times 3 = 6}} - \frac{x^{\times 2}}{3^{\times 2 = 6}} = \frac{1^{\times 5}}{4_{\times 5 = 20}} + \frac{1^{\times 20}}{1_{\times 20 = 20}} + \frac{1^{\times 4}}{5_{\times 4 = 20}}$$

LCM of 2, 3 = 6 LCM of 1, 4, 5 = 20

$$\frac{3x - 2x}{6} = \frac{5 + 20 + 4}{20}$$

$$\frac{x}{6} = \frac{29}{20}$$

On shifting 6 to RHS side it will be multiplied

$$x = \frac{29 \times \cancel{6}^{3}}{\cancel{20}_{10}}$$

$$x = \frac{82}{10} \quad \text{OR} \quad 8.2$$

(b) $\frac{x+1}{2x+3} = \frac{3}{8}$

Shift 8 to LHS and $(2x+3)$ to RHS they both are in divide on shifting they will be multiplied.

$$8(x+1) = 3(2x+3)$$

$$8x+8 = 6x+9$$

$$8x-6x = 9-8$$

$$2x = 1$$

$$x = \frac{1}{2} \quad \text{OR} \quad 0.5$$